MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL, TRANSPARENT AND OCTAL D TYPE LATCHES, CASCADABLE, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Advanced Schottky TTL, Octal transparent and Octal D type latches (three state) microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Octal transparent latch with active low enable 3-state outputs
02	Octal transparent latch with active low enable 3-state outputs
03	Octal D-type latch with 3-state inverted outputs
04	Octal D-type latch with 3 state outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
R	D-8 (20 lead, 1/4" x 1-1/16"), dual-in-line package
S	F-9 (20 lead, 1/4" x 1/2"), flat package
2	C-2 (20 terminal, .350" x .350"), square chip carrier

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D): 1/	
Device type 01	330 mW
Device type 02	385 mW
Device type 03	336 mW
Device type 04	303 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases R,S	(See MIL-M-38510, appendix C)
Case 2	60°C/W 2/
Junction temperature (T_J) 3/	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).
2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value state herein.
3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 V dc
Width of latch enable pulse high t_{pw} (LEH):	
Device type 01	5.0 ns minimum
Device type 02	6.0 ns minimum
Device types 03, 04	4.0 ns minimum
Setup time data high to latch enable pulse T_S (H):	
Device type 01	2.0 ns minimum
Device types 02, 03, 04	2.0 ns minimum
Setup time data low to latch enable pulse T_S (L):	
Device type 01	2.0 ns minimum
Device types 02, 03, 04	2.0 ns minimum
Hold time data high to latch enable pulse T_H (H):	
Device type 01	4.0 ns minimum
Device types 02, 03, 04	3.0 ns minimum
Hold time data low to high latch enable pulse T_H (L):	
Device type 01	4.0 ns minimum
Device type 02	3.0 ns minimum
Device type 03	3.0 ns minimum
Device type 04	4.0 ns minimum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

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TABLE I. Electrical performance characteristics.

	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -1.0 mA V_{IL} = 0.8 V, V_{IH} = 2.0 V	All	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OH} = 20 mA V_{IL} = 0.8 V, V_{IH} = 2.0 V	All		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = +25°C	All		-1.2	V
High-level input current	I_{IH1} I_{IH2}	V_{CC} = 5.5 V, V_{IH} = 2.7 V V_{CC} = 5.5 V, V_{IH} = 7.0 V	All		20 100	uA uA
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IL} = 0.5 V	All	-0.03	-.60	mA
Short-circuit output current 1/	I_{OS}	V_{CC} = 5.5 V, V_{OS} = 0.0 V	All	-60	-150	mA
Output drive	I_{OD}	V_{CC} = 4.5 V, V_{IN} = 2.5 V V_{OUT} = 2.5 V	01,02	35		mA
High level supply current	I_{CCH}	V_{CC} = 5.5 V	01 02		55 50	mA mA
Low level supply current	I_{CCL}	V_{CC} = 5.5 V	01 02 03 04		60 70 61 55	mA mA mA mA
Off-state supply current	I_{CCZ}	V_{CC} = 5.5 V Outputs disabled	01 02,03 04		60 61 55	mA mA mA
Off-state output leakage current	I_{OZH}	V_{CC} = 5.5 V, V_{ZH} = 2.7 V	All		50	uA
	I_{OZL}	V_{CC} = 5.5 V, V_{ZL} = 0.5 V	All		-50	uA
Propagation delay time, low-to-high level		V_{CC} = 5.0 V (see figure 5) C_L = 50 pF ±10%				
D to Q	t_{PLH1}		01 04	3.0 3.0	8.5 9.0	ns ns
D to $\overline{Q}$	t_{PLH1}		02 03	4.0 3.0	12.0 10.5	ns ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
LE to Q	t_{PLH2}		01	5.0	15.5	ns
			04	5.0	13.5	ns
LE to $\overline{Q}$	t_{PLH2}		02	4.0	14.0	ns
			03	4.0	11.0	ns
Propagation delay time, high-to-low level						
D to Q	t_{PHL1}		01	2.0	7.0	ns
			04	2.0	7.0	ns
D to $\overline{Q}$	t_{PHL1}		02	3.0	9.0	ns
			03	2.0	7.5	ns
LE to Q	t_{PHL2}		01	3.0	8.5	ns
			04	3.0	7.5	ns
LE to $\overline{Q}$	t_{PHL2}		02	3.0	9.0	ns
			03	2.5	7.5	ns
Propagation delay time, low level to off-state						
$\overline{OE}$ to Q	t_{PLZ1}		01	1.5	7.0	ns
			04	1.5	5.5	ns
$\overline{OE}$ to $\overline{Q}$	t_{PLZ1}		02	2.0	7.5	ns
			03	1.5	5.5	ns
Propagation delay time, high level to off-state						
$\overline{OE}$ to Q	t_{PHZ1}		01	1.5	10.0	ns
			04	1.5	7.0	ns
$\overline{OE}$ to $\overline{Q}$	t_{PHZ1}		02	2.0	8.5	ns
			03	1.5	7.0	ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
Propagation delay time, off-state to low level		V_{CC} = 5.0 V (See figure 5) C_L = 50 pF ±10%				
$\overline{OE}$ to Q	t_{PZL1}		01 04	2.0 2.0	10.0 10.0	ns ns
$\overline{OE}$ to $\overline{Q}$	t_{PZL1}		02 03	2.0 2.5	9.0 10.0	ns ns
Propagation delay time, off-state to high level						
$\overline{OE}$ to Q	t_{PZH1}		01 04	2.0 2.0	15.0 10.0	ns ns
$\overline{OE}$ to $\overline{Q}$	t_{PZH1}		02 03	2.0 2.0	12.5 9.5	ns ns

1/ Not more than one output should be shorted at a time.

3.2.1 Case outlines. The case outlines shall be as specified in 1.2.3.

3.2.2 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figures 1 and 2.

3.2.3 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 3.

3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 10 (see MIL-M-38510, appendix E).

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,8,9
Group B test requirements (method 5005) subgroup 5	1,2,3,7,8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D, E, or F using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Test shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to group C inspection parameters for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

c. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, E, or F using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

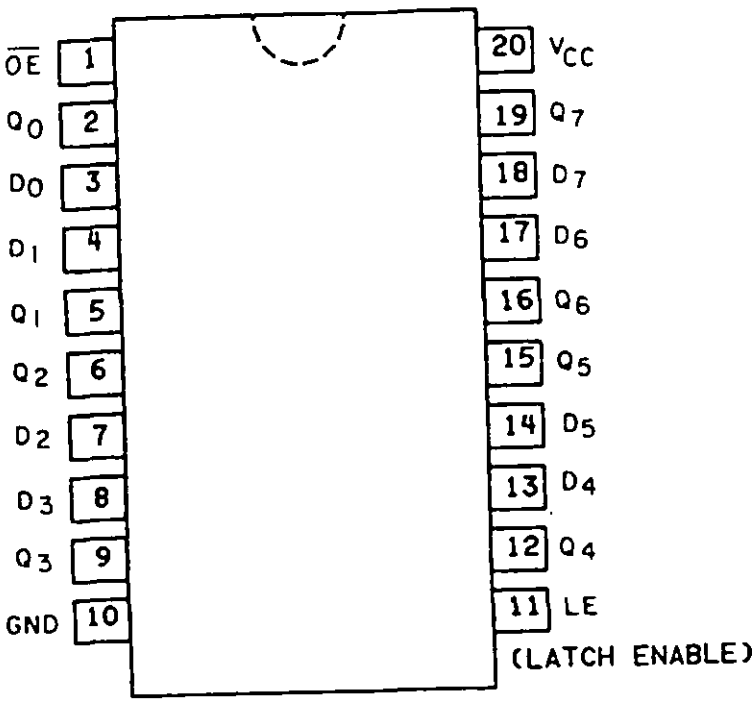
b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

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Device type 01
Cases R and S



Device type 02
Cases R and S

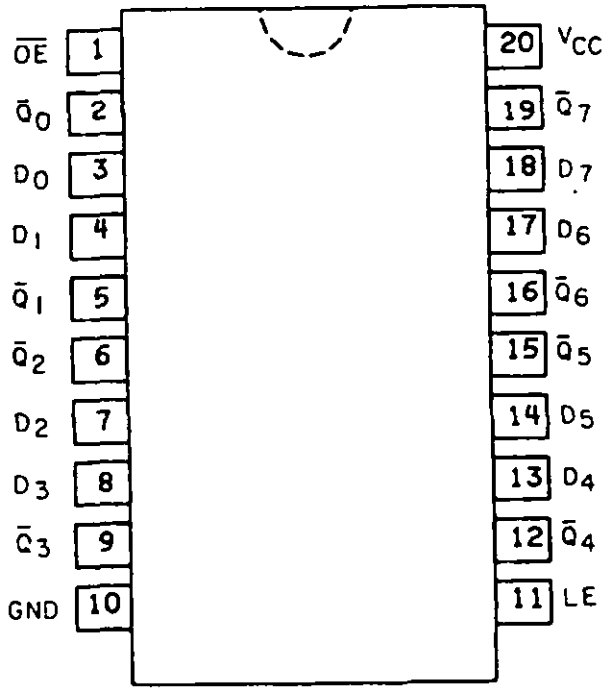
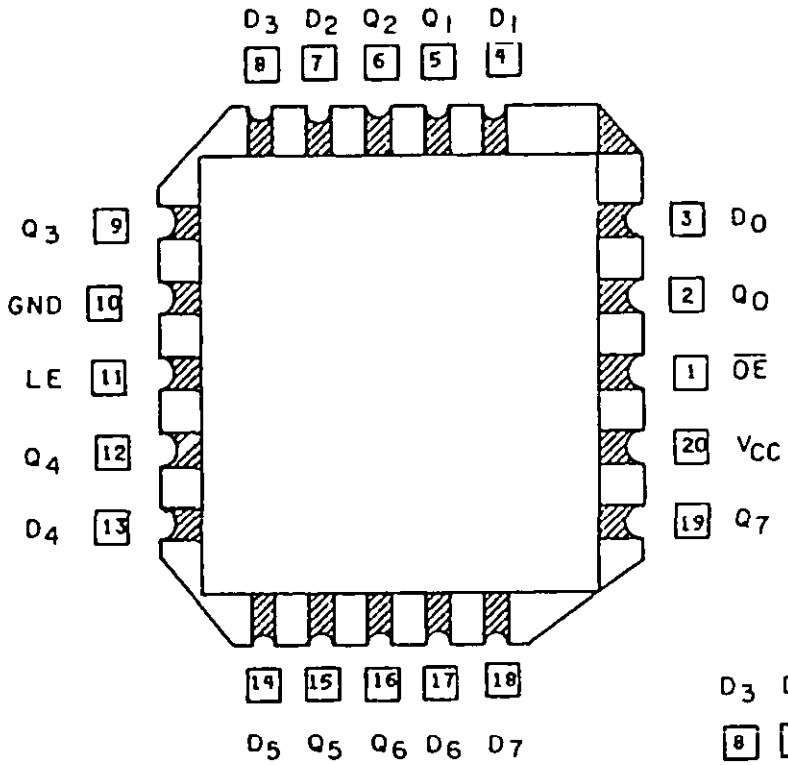


FIGURE 1. Terminal connections.

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Device type 01

Case 2



Device type 02

Case 2

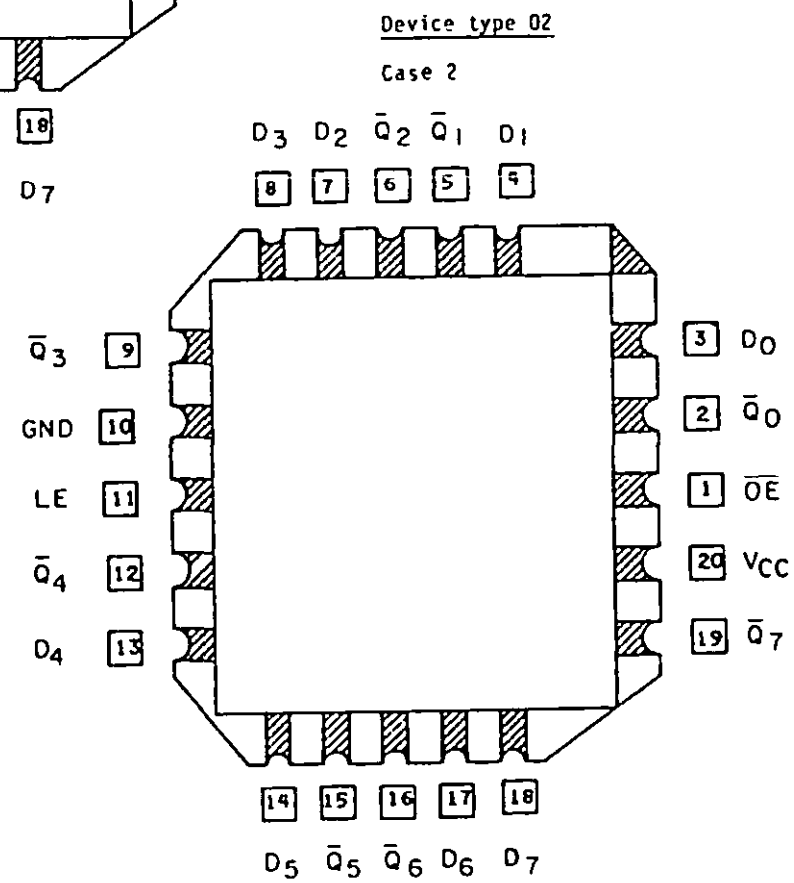
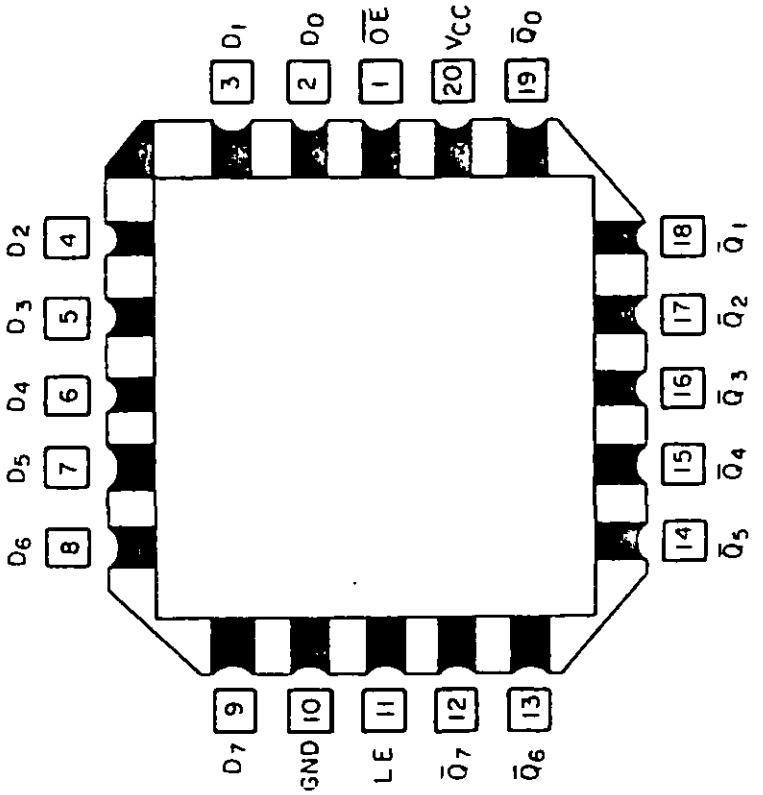


FIGURE 1. Terminal connections - Continued.

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Device type 03

Case 2



Device type 03

Cases R and S

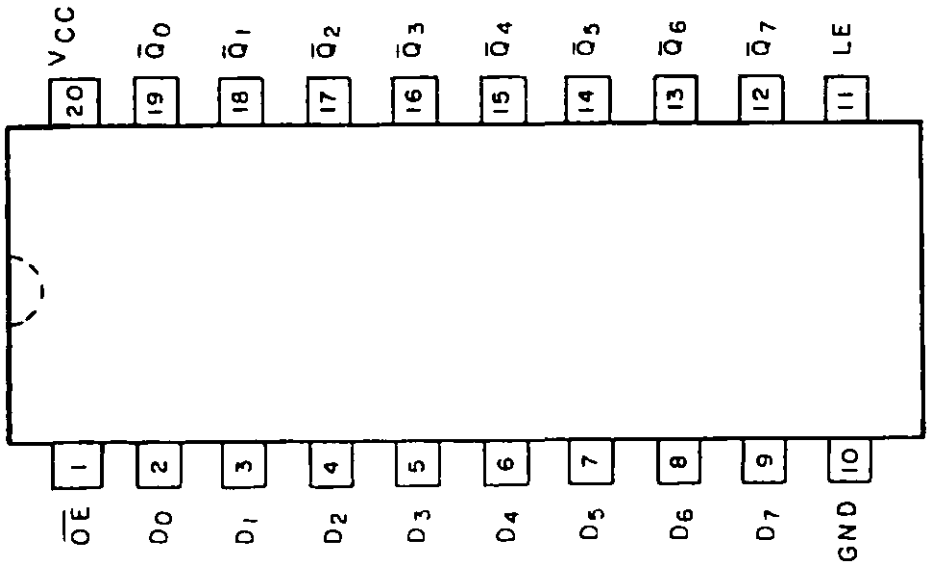
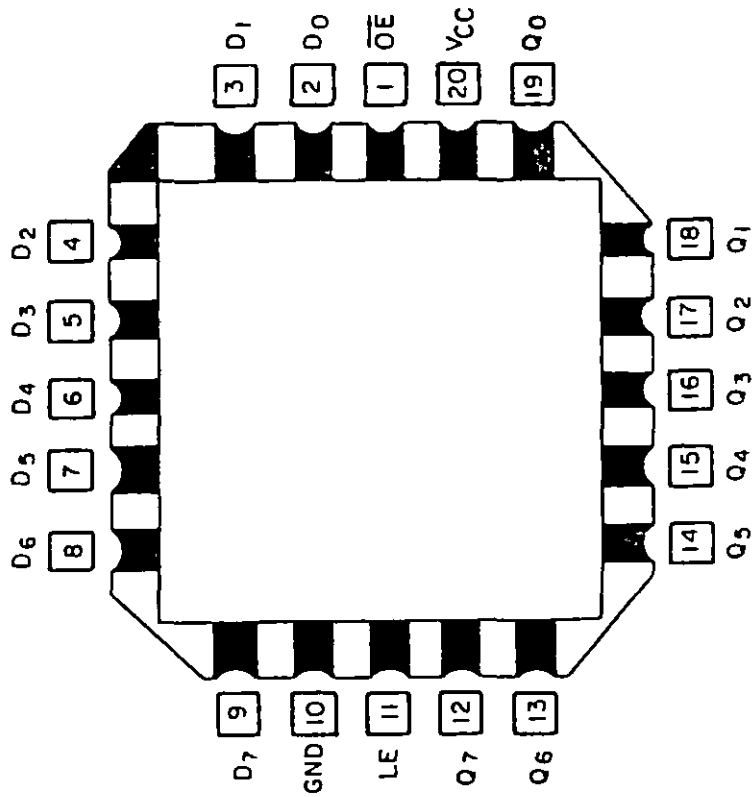


FIGURE 1. Terminal connections. (Top view) - Continued.

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Device type 04

Case 2



Device type 04

Cases R and S

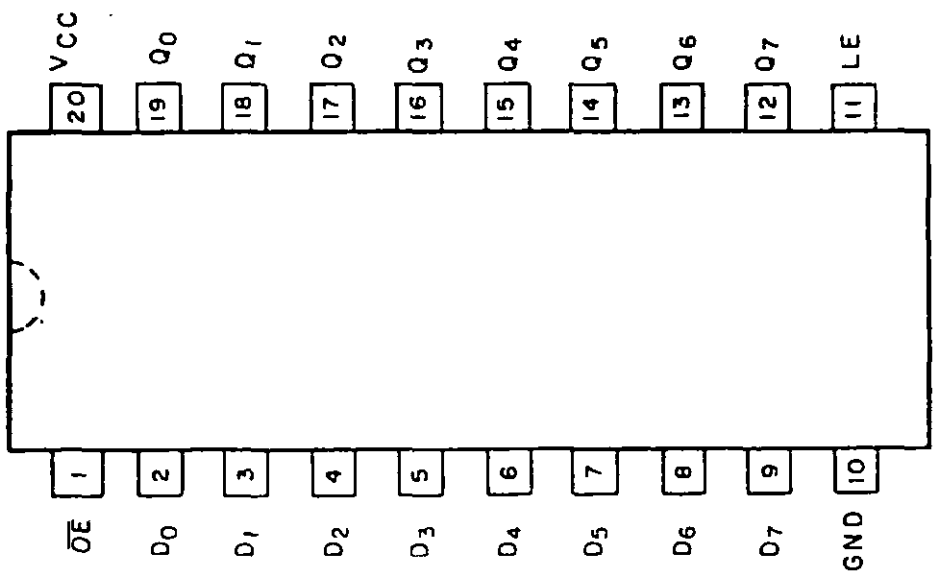
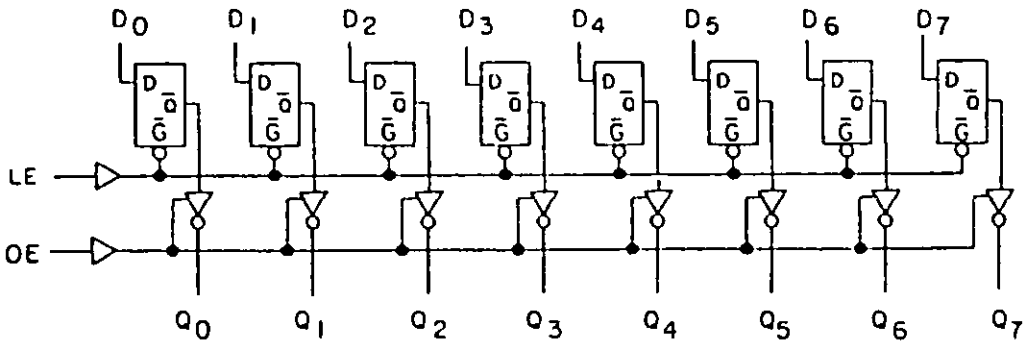
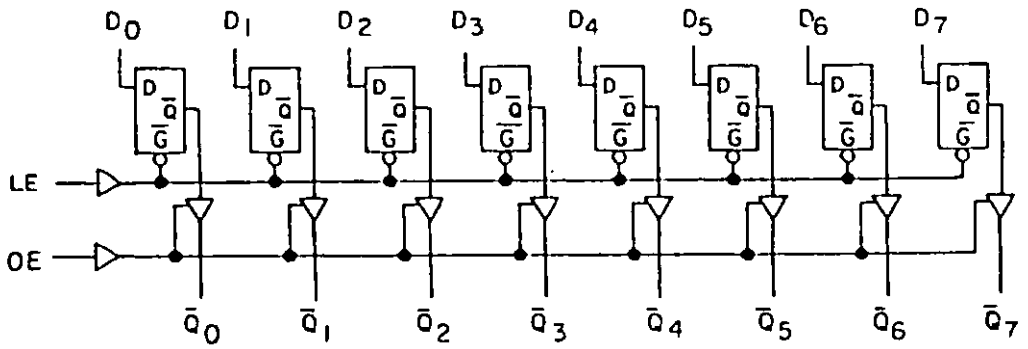


FIGURE 1. Terminal connections. (Top view) -Continued.

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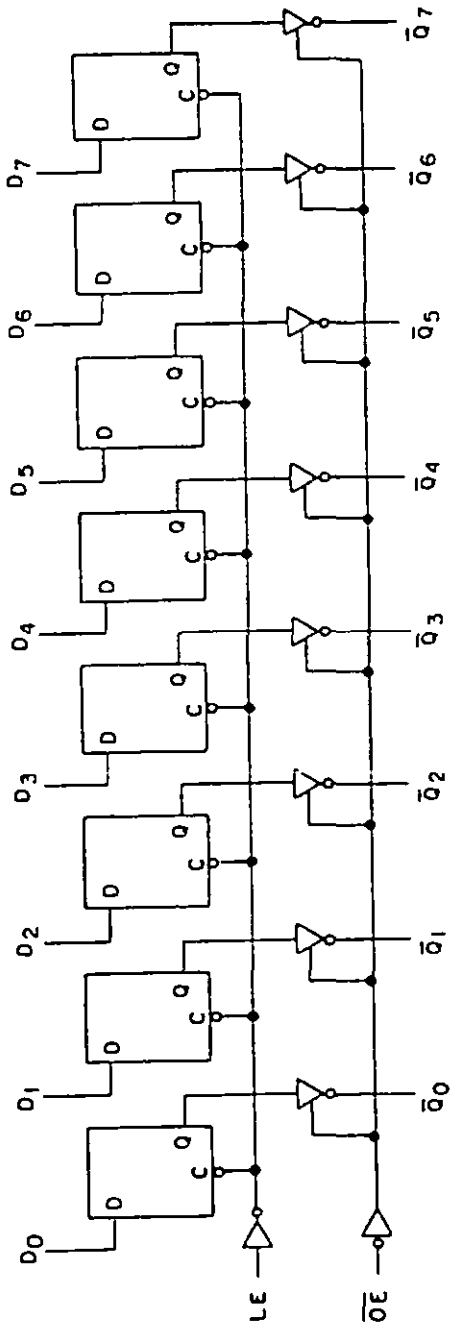
Device type 01



Device type 02

FIGURE 2. Logic diagrams.

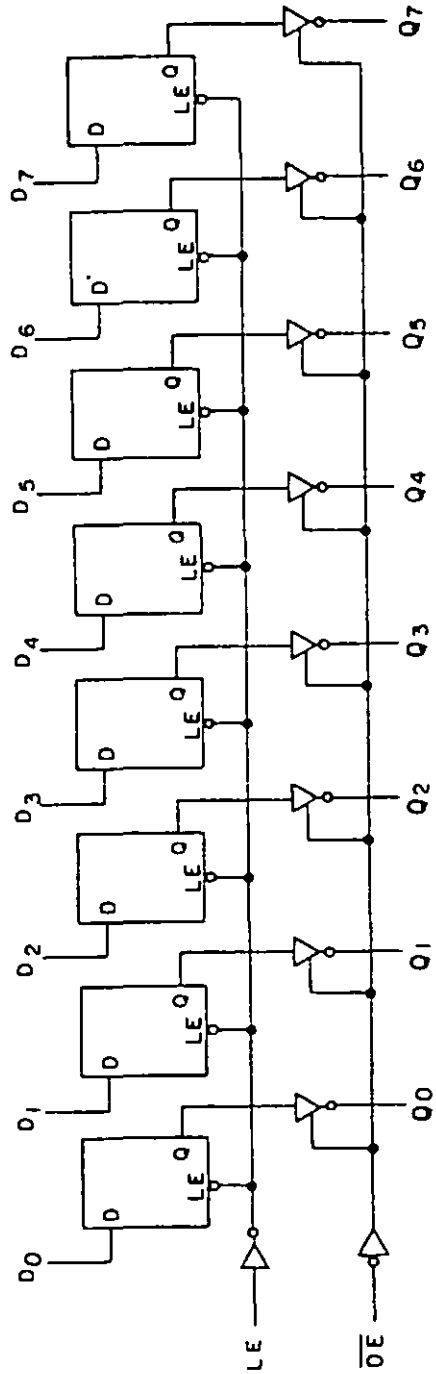
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Device type 03

FIGURE 2. Logic diagrams - Continued.

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Device type 04

FIGURE 2. Logic diagrams - Continued.

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Device types 01 and 04

Inputs				Outputs	
OE	Dn	LE		Qn	
		t	t+1	t	t+1
H	X	X	X	Z	Z
L	H	H	H	H	H
L	L	H	H	L	L
L	H	H	L	H	H
L	L	H	L	L	L
L	X	L	L	Q0	Q0
L	H	L	H	Q0	H
L	L	L	H	Q0	L

Device type 02

Inputs				Outputs	
OE	Dn	LE		Qn	
		t	t+1	t	t+1
H	X	X	X	Z	Z
L	H	H	H	L	L
L	L	H	H	H	H
L	H	H	L	L	L
L	L	H	L	H	H
L	X	L	L	Q0	Q0
L	H	L	H	Q0	L
L	L	L	H	Q0	H

H = HIGH voltage level
L = LOW voltage level
X = Immaterial
Z = High Impedance
Q0 = output prior to last H to L transition of LE

FIGURE 3. Truth tables.

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Inputs			Internal	Output	Function
$\overline{OE}$	LE	D	Q	O	
H	X	X	X	Z	High Z
H	H	L	H	Z	High Z
H	H	H	L	Z	High Z
H	L	X	NC	Z	Latched
L	H	L	H	H	Transparent
L	H	H	L	L	Transparent
L	L	X	NC	NC	Latched

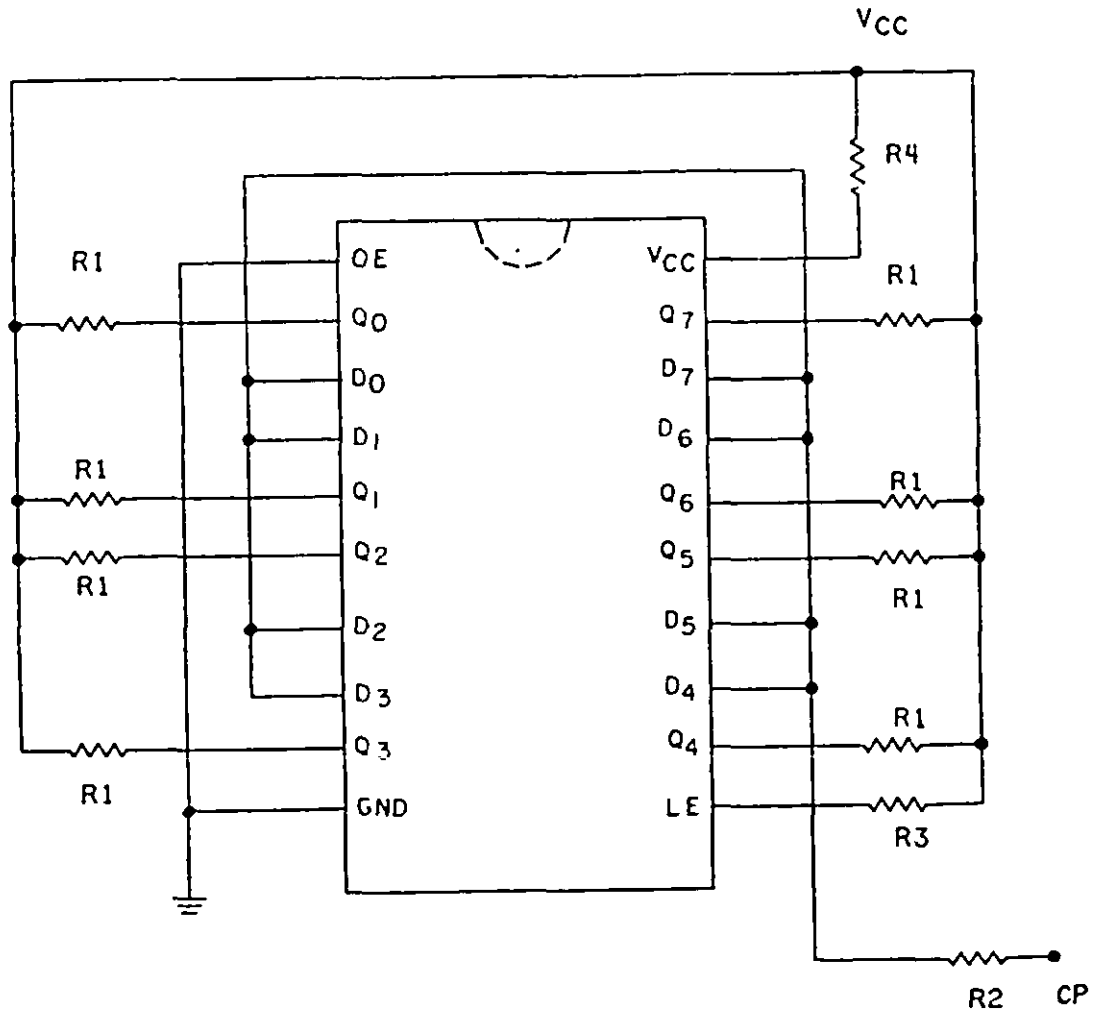
Device type 03

H - HIGH voltage level
L - LOW voltage level
X - Immaterial.
Z - High Impedance
NC - No change

FIGURE 3. Truth tables -Continued.

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Device type 01

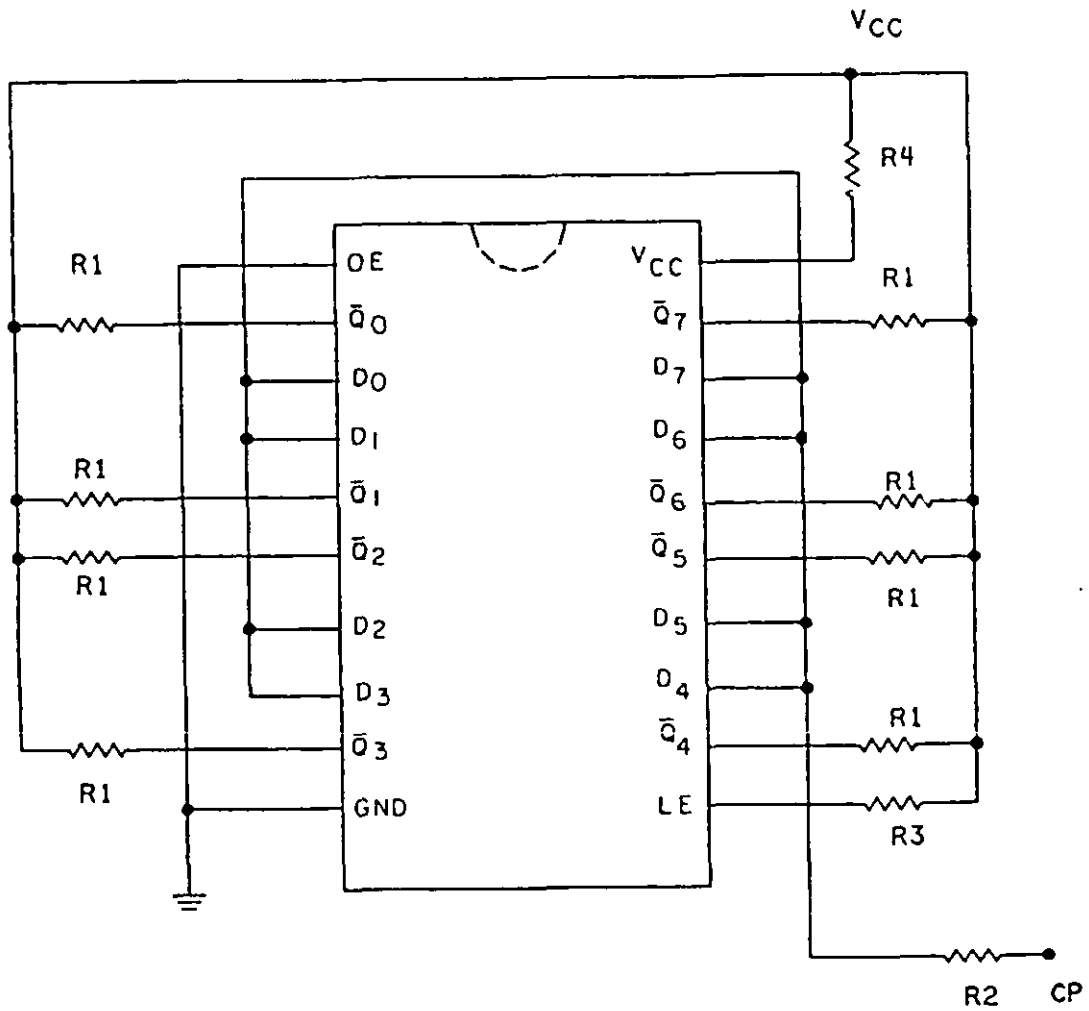


NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω ±5%; R2 = 51 kΩ ±5%; R3 = 1 kΩ ±5%.
3. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit.

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Device type 02

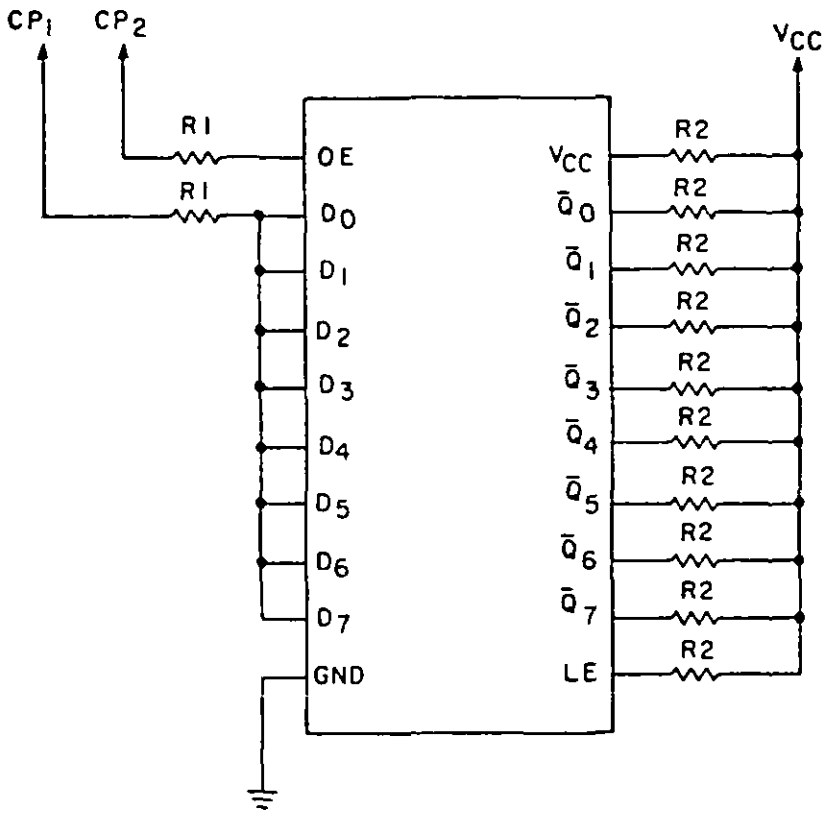


NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω ±5%; R2 = 51 kΩ ±5%; R3 = 1 kΩ ±5%.
3. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit.

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Device type 03

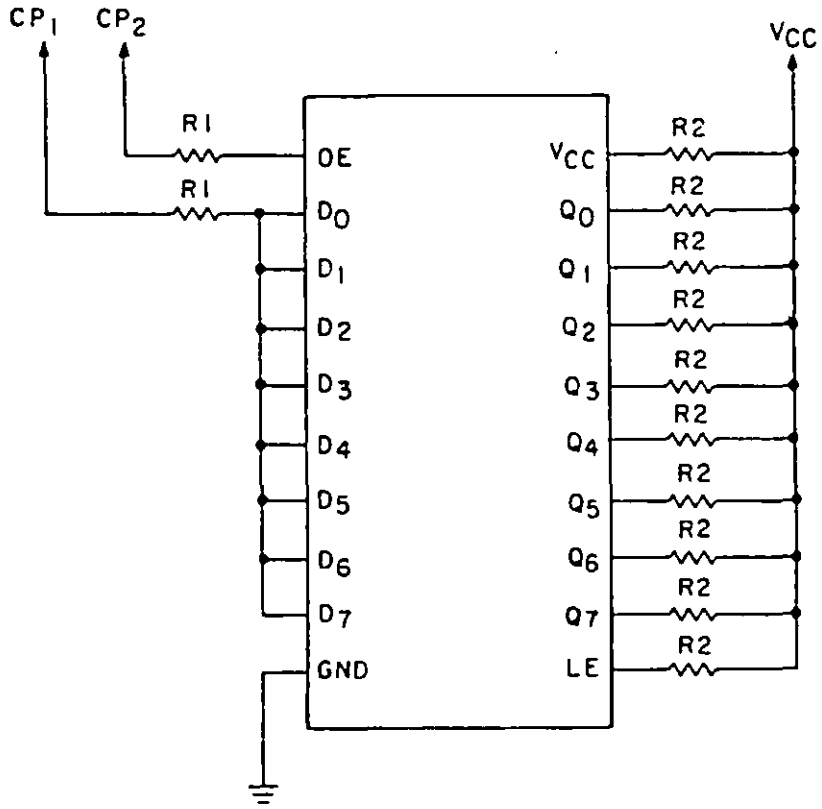


NOTES:
1. CP_1 = 100 kHz ±50% square wave; duty cycle = 50 ±15%; CP_2 = CP_1 ÷ 2; V_{IL} = -0.5 minimum to 0.9 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R_1 = 51Ω ±5%, R2 = 210Ω maximum.
3. V_{CC} and R3 shall be chosen to insure 5.5 minimum is present at device V_{CC}.

FIGURE 4. Burn-in and life test circuit

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Device type 04



NOTES:
1. CP_1 = 100 kHz ±50% square wave; duty cycle = 50 ±15%; CP_2 = CP_1 - 2; V_{IL} = -0.5 minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R_1 = 51Ω ±5%, R2 = 240Ω maximum.
3. V_{CC} and R3 shall be chosen to insure 5.5 minimum is present at device V_{CC}.

FIGURE 4. Burn-in and life test circuit.

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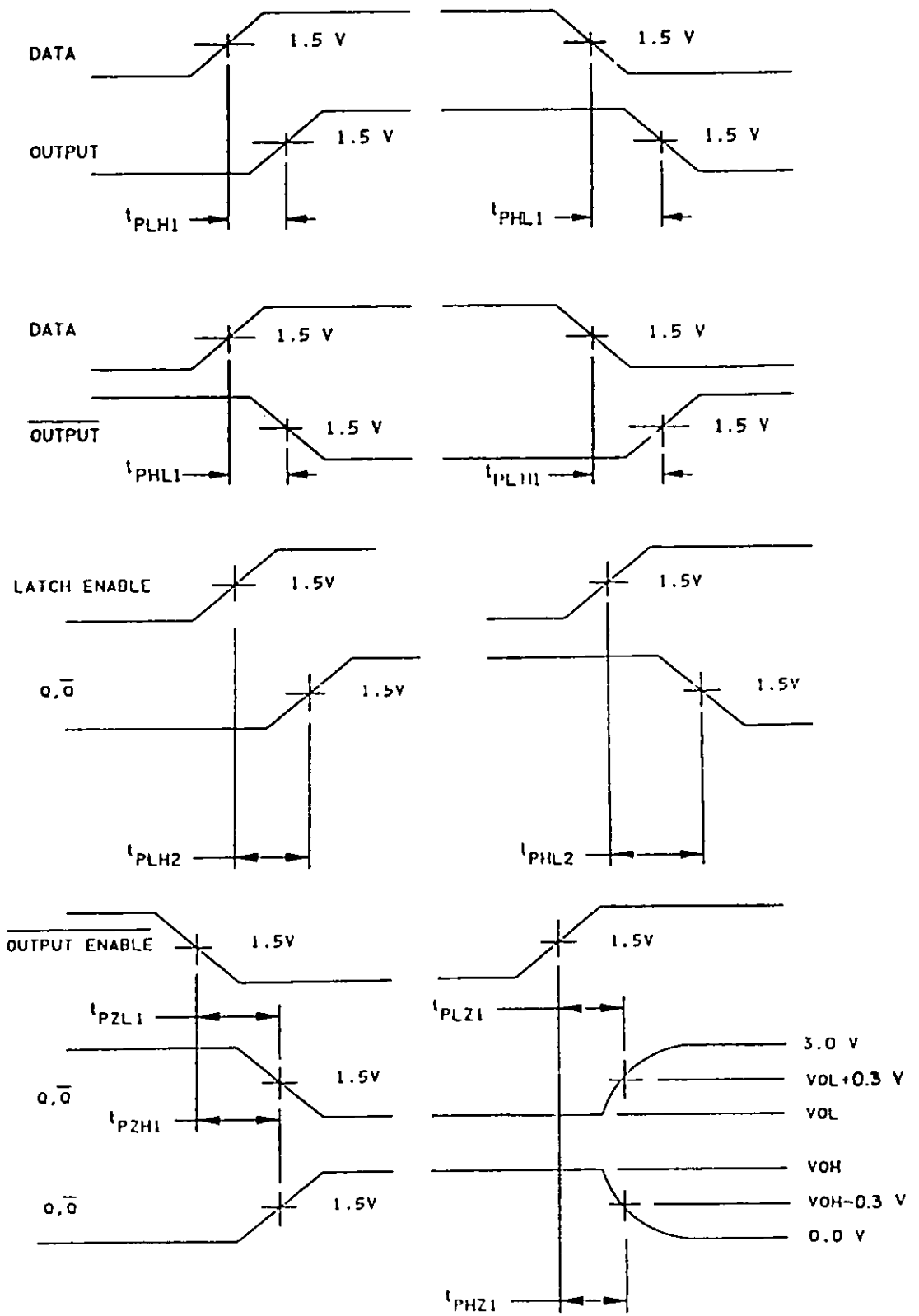
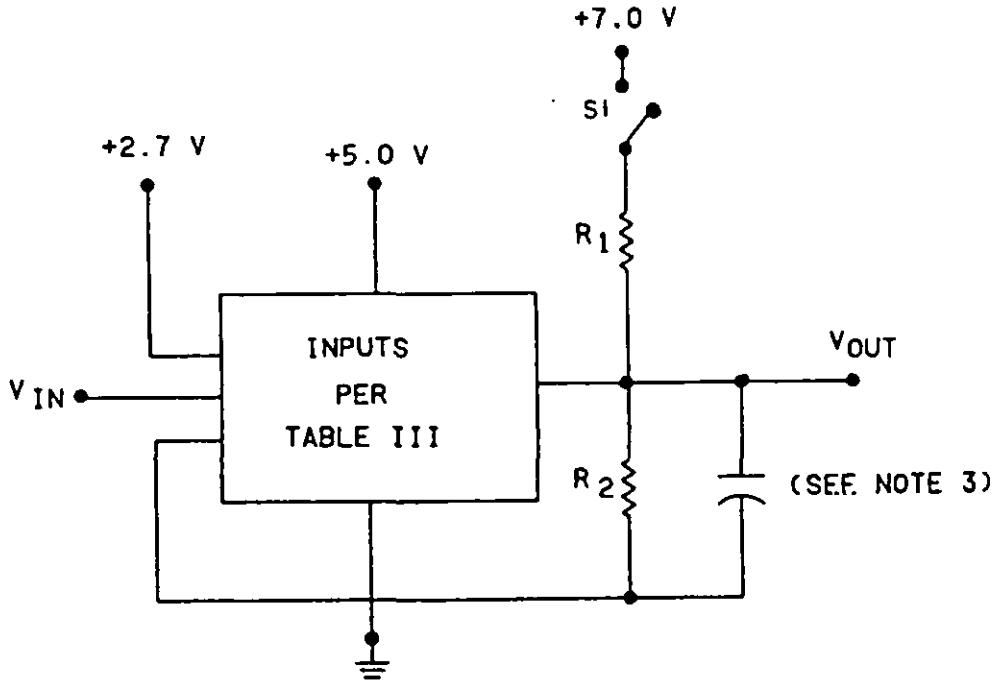
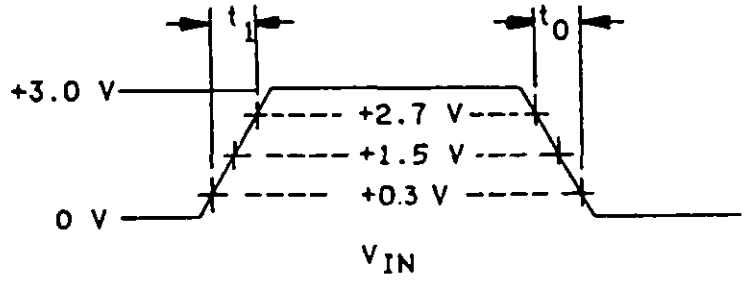


FIGURE 5. Switching time waveforms.

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TEST TYPE	S1
t_{PLH}	OPEN
t_{PHL}	OPEN
t_{PHZ}	OPEN
t_{PZH}	OPEN
t_{PLZ}	CLOSED
t_{PZL}	CLOSED



NOTES:
1. Input pulse has the following characteristics: $t_1 = t_0 \leq 2.5$ ns, PRR $\leq$ 1 MHz, $Z_{OUT} \cong$ 50Ω.
2. Inputs not under test are at GND.
3. C_L = 50 pF ±10% including scope, probe, wiring, and stray capacitance, without package in test fixture.
4. R_1 and R_2 = 499Ω ±5%.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Switching time test circuit - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1,5,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OE}$	O_0	D_0	D_1	O_1	O_2	D_2	D_3	O_3	GND	LE	D_4	O_4	D_5	O_5	O_6	D_6	D_7	O_7	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	0.8 V	-1 mA	2.0 V	2.0 V	-1 mA	-1 mA	2.0 V	2.0 V	-1 mA	GND	2.0 V	-1 mA	2.0 V	2.0 V	-1 mA	-1 mA	2.0 V	2.0 V	-1 mA	4.5 V	O_0 O_1 O_2 O_3 O_4 O_5 O_6 O_7	2.5		V
	V_{OL}	3007	9		20 mA	.8 V	.4 V	20 mA	20 mA	.8 V	.8 V	20 mA			20 mA	.8 V	.8 V	20 mA	20 mA	.8 V	.8 V	20 mA		O_0 O_1 O_2 O_3 O_4 O_5 O_6 O_7		0.5	
	V_{IC}		17	-18 mA		-18 mA	-18 mA			-18 mA	-18 mA			-18 mA		-18 mA	-18 mA			-18 mA	-18 mA			$\overline{OE}$ D_0 D_1 D_2 D_3 O_4 O_5 O_6 O_7		-1.2	
	I_{IH1}	3010	27	2.7 V		2.7 V	2.7 V			2.7 V	2.7 V			2.7 V		2.7 V	2.7 V			2.7 V	2.7 V		5.5 V	$\overline{OE}$ D_0 D_1 D_2 D_3 O_4 O_5 O_6 O_7		20	uA
	I_{IH2}		37	7.0 V		7.0 V	7.0 V			7.0 V	7.0 V			7.0 V		7.0 V	7.0 V			7.0 V	7.0 V			$\overline{OE}$ D_0 D_1 D_2 D_3 O_4 O_5 O_6 O_7		100	

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases B, C, 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OE}$	O_3	O_0	O_1	O_1	O_2	O_2	O_3	O_3	GND	LE	O_4	O_4	O_5	O_5	O_6	O_6	O_7	O_7	V_{CC}				
1 T_C = +25°C	I_{IL}	3009	47	0.5 V		0.5 V							GND										5.5 V	$\overline{OE}$	4/	4/	mA
			48				0.5 V																	0			
			49							0.5 V				0.5 V										1			
			50								0.5 V					0.5 V								2			
			51														0.5 V							3			
			52															0.5 V						4			
			53																0.5 V					5			
			54																	0.5 V				6			
			55																		0.5 V			7			
			56																								
	I_{OZH}		57	2.0 V	2.4 V	0.0 V	0.0 V	2.4 V	2.4 V	0.0 V	3.0 V	2.4 V		5.5 V	2.4 V	0.0 V	0.0 V	2.4 V	2.4 V	0.0 V	0.0 V	2.4 V		0		50	µA
			58				0.0 V																	1			
			59						2.4 V	0.0 V						0.0 V								2			
			60								3.0 V	2.4 V						2.4 V						3			
			61																2.4 V	0.0 V				4			
			62																		0.0 V	2.4 V		5			
			63																					6			
			64																					7			
	I_{OZL}		65	0.5 V	0.5 V	5.5 V	5.5 V	0.5 V	0.5 V	5.5 V	5.5 V	0.5 V			0.5 V	5.5 V	5.5 V	0.5 V	0.5 V	5.5 V	5.5 V	0.5 V		0		-50	µA
			66				0.5 V																	1			
			67						0.5 V	5.5 V						5.5 V								2			
			68								5.5 V	0.5 V						0.5 V						3			
			69																0.5 V	5.5 V				4			
			70																		5.5 V	0.5 V		5			
			71																					6			
			72																					7			
	I_{OS}	3011	73	0.0 V	0.0 V	4.5 V	4.5 V	0.0 V	0.0 V	4.5 V	4.5 V	0.0 V			0.0 V	4.5 V	4.5 V	0.0 V	0.0 V	4.5 V	4.5 V	0.0 V		0	-60	-150	mA
			74				4.5 V																	1			
			75						0.0 V	4.5 V						4.5 V								2			
			76								4.5 V	0.0 V						0.0 V						3			
			77																0.0 V	4.5 V				4			
			78																		4.5 V	0.0 V		5			
			79																					6			
			80																					7			
	I_{OD}		81	0.0 V	2.5 V	0.0 V	0.0 V	2.5 V	2.5 V	0.0 V	0.0 V	2.5 V			2.5 V	0.0 V	0.0 V	2.5 V	2.5 V	0.0 V	0.0 V	2.5 V	4.5 V	0	33		mA
			82				0.0 V																	1			
			83						2.5 V	0.0 V						0.0 V								2			
			84								0.0 V	2.5 V						2.5 V						3			
			85																2.5 V	0.0 V				4			
			86																		0.0 V	2.5 V		5			
			87																					6			
			88																					7			
	I_{CCL}	3005	89	0.0 V		0.0 V	0.0 V			0.0 V	0.0 V		*	*		0.0 V	0.0 V			0.0 V	0.0 V		5.5 V	V_{CC}		60	*
	I_{CCH}	*	90	0.0 V		5.5 V	5.5 V			5.5 V	5.5 V		*	*		5.5 V	5.5 V			5.5 V	5.5 V		*	*		55	*
	I_{CC2}	*	91	5.5 V									*	*									*	*		60	*
2 T_C = +125°C	Same tests as subgroup 1 except V_{IC} tests are omitted.																										
3 T_C = -55°C	Same tests as subgroup 1 except V_{IC} tests are omitted.																										

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 Method	Cases R,S,Z Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OE}$	O_0	O_0	D_1	O_1	D_2	D_2	D_3	O_3	GND	LE	O_4	O_4	O_5	O_5	O_6	O_6	O_7	O_7	V_{CC}				
7 T_C = +25°C	Functional tests 1/2/3/	3014	92 93 94 95 96 97 98 99 100	B " " " " " " " "	L H " " L H " " "	B A A B B A B B A	A B B A A B A A B	H L " " H L " " "	L H " " L H " " "	B A A B B A B B A	A B B A A B A A B	H L " " H L " " "	GND " " " " " " " "	A A B B A " " " B	L H " " L " " " "	B A A B B A B B A	A B B A A B A A B	H L " " H L " " "	L H " " L H " " "	B A A B B A B B A	A B B A A B A A B	H L " " H L " " "	3/ " " " " " " " "	All outputs " " " " " " " "			
8	Repeat subgroup 7 tests at T_C = +125°C and T_C = -55°C.																										
9 T_C = +25°C	t_{PLH1}	3003 Fig. 5	101 102 103 104 105 106 107 108	0.0 V " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	GND " " " " " " "	2.7 V " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	5.0 V " " " " " " "	$I0_0$ to O_0 $I0_1$ to O_1 $I0_2$ to O_2 $I0_3$ to O_3 $I0_4$ to O_4 $I0_5$ to O_5 $I0_6$ to O_6 $I0_7$ to O_7	3.0 " " " " " " "	7.0 " " " " " " "	ns " " " " " " "
	t_{PHL1}	" " " " " " " "	109 110 111 112 113 114 115 116	" " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	" " " " " " " "	" " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	IN " " " " " " "	IN " " " " " " "	OUT " " " " " " "	" " " " " " " "	$I0_0$ to O_0 $I0_1$ to O_1 $I0_2$ to O_2 $I0_3$ to O_3 $I0_4$ to O_4 $I0_5$ to O_5 $I0_6$ to O_6 $I0_7$ to O_7	2.0 " " " " " " "	5.0 " " " " " " "	" " " " " " " "
	t_{PLH2}	" " " " " " " "	117 118 119 120 121 122 123 124	" " " " " " " "	OUT " " " " " " "	2.7 V " " " " " " "	2.7 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	2.7 V " " " " " " "	2.7 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	IN " " " " " " "	OUT " " " " " " "	2.7 V " " " " " " "	2.7 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	2.7 V " " " " " " "	2.7 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	LE to O_0 LE to O_1 LE to O_2 LE to O_3 LE to O_4 LE to O_5 LE to O_6 LE to O_7	5.0 " " " " " " "	11.5 " " " " " " "	" " " " " " " "
	t_{PHL2}	" " " " " " " "	125 126 127 128 129 130 131 132	" " " " " " " "	OUT " " " " " " "	0.0 V " " " " " " "	0.0 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	0.0 V " " " " " " "	0.0 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	" " " " " " " "	OUT " " " " " " "	0.0 V " " " " " " "	0.0 V " " " " " " "	OUT " " " " " " "	OUT " " " " " " "	0.0 V " " " " " " "	0.0 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	LE to O_0 LE to O_1 LE to O_2 LE to O_3 LE to O_4 LE to O_5 LE to O_6 LE to O_7	3.0 " " " " " " "	7.0 " " " " " " "	" " " " " " " "

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A, S, Z test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OE}$	I_3	O_0	I_1	O_1	O_2	I_2	O_3	I_3	GND	LE	O_4	I_4	I_5	O_5	O_6	I_6	O_7	I_7	V_{CC}			
9 T_C = +25°C	t_{PHZL}	3003 (Fig. 5	1133 1134 1135 1136 1137 1138 1139 1140	In	OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT	GND	2.7 V	OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT	5.0 V	1 to 8	1.5	6.5	ns
	t_{PLZL}		1141 1142 1143 1144 1145 1146 1147 1148		OUT	0.0 V	0.0 V	OUT	OUT	0.0 V	0.0 V	OUT			OUT	0.0 V	0.0 V	OUT	OUT	0.0 V	0.0 V	OUT		1 to 8			
	t_{PZHL}		1149 1150 1151 1152 1153 1154 1155 1156		OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT			OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT	5.0 V	1 to 8	2.0	11.0	
	t_{PZLL}		1157 1158 1159 1160 1161 1162 1163 1164		OUT	0.0 V	0.0 V	OUT	OUT	0.0 V	0.0 V	OUT			OUT	0.0 V	0.0 V	OUT	OUT	0.0 V	0.0 V	OUT		1 to 8		7.5	
10 T_C = +125°C	Same tests as subgroup 9 except use limits from table I.																										
11 T_C = -55°C	Same tests as subgroup 10 except T_C = -55°C.																										

1/ A = 2.5 V minimum, B = 0.5 V or GND.

2/ H ≥ 1.5 V, L ≤ 1.5 V

3/ Perform subgroup 7 sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

4/ I_{IL} limits (mA) min/max values for circuit shown:

Parameter	Circuits A	Circuits B	Circuits C
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A, S, 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OE}$	$\overline{O_0}$	O_0	O_1	$\overline{O_1}$	$\overline{O_2}$	O_2	O_3	$\overline{O_3}$	GND	LE	$\overline{O_4}$	O_4	O_5	$\overline{O_5}$	$\overline{O_6}$	O_6	O_7	$\overline{O_7}$	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1 2 3 4 5 6 7 8	0.8 V	-1 mA	0.8 V	0.8 V	-1 mA	-1 mA	0.8 V	0.8 V	-1 mA	GND	2.0 V	-1 mA	.8 V	.8 V	-1 mA	-1 mA	.8 V	.8 V	-1 mA	4.5 V		2.5		V
	V_{OL}	3007	9 10 11 12 13 14 15 16		20 mA	2.0 V	2.0 V	20 mA	20 mA	2.0 V	2.0 V	20 mA			20 mA	2.0 V	2.0 V	20 mA	20 mA	2.0 V	2.0 V	20 mA				0.5	
	V_{IC}		17 18 19 20 21 22 23 24 25 26	-18 mA		-18 mA	-18 mA			-18 mA	-18 mA			-18 mA		-18 mA	-18 mA			-18 mA	-18 mA					-1.2	
	I_{IH1}	3010	27 28 29 30 31 32 33 34 35 36	2.7 V		2.7 V	2.7 V			2.7 V	2.7 V			2.7 V		2.7 V	2.7 V			2.7 V	2.7 V		5.5 V			20	µA
	I_{IH2}	3010	37 38 39 40 41 42 43 44 45 46	7.0 V		7.0 V	7.0 V			7.0 V	7.0 V			7.0 V		7.0 V	7.0 V			7.0 V	7.0 V					100	

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,S,2 Test no.	1 $\overline{OE}$	2 $\overline{O_0}$	3 D_0	4 O_1	5 $\overline{O_1}$	6 $\overline{O_2}$	7 O_2	8 O_3	9 $\overline{O_3}$	10 GND	11 I.E	12 $\overline{O_4}$	13 O_4	14 O_5	15 $\overline{O_5}$	16 $\overline{O_6}$	17 O_6	18 O_7	19 $\overline{O_7}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IL}	3009	47 48 49 50 51 52 53 54 55 56	0.5 V		0.5 V	0.5 V			0.5 V	0.5 V		GND	0.5 V		0.5 V	0.5 V			0.5 V	0.5 V		5.5 V		4/	4/	mA
	I_{OZH}		57 58 59 60 61 62 63 64	2.0 V	2.4 V	5.5 V	5.5 V	2.4 V	2.4 V	5.5 V	5.5 V	2.4 V		5.5 V	2.4 V	5.5 V	5.5 V	2.4 V	2.4 V	5.5 V	5.5 V	2.4 V				50	µA
	I_{OZL}		65 66 67 68 69 70 71 72		0.5 V	0.0 V	0.0 V	0.5 V	0.5 V	0.0 V	0.0 V	0.5 V			0.5 V	0.0 V	0.0 V	0.5 V	0.5 V	0.0 V	0.0 V	0.5 V				-50	
	I_{OS}	3011	73 74 75 76 77 78 79 80	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V			0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V			-60	-150	mA
	I_{OD}		81 82 83 84 85 86 87 88	0.0 V	2.5 V	5.5 V	5.5 V	2.5 V	2.5 V	5.5 V	5.5 V	2.5 V			2.5 V	5.5 V	5.5 V	2.5 V	2.5 V	5.5 V	5.5 V	2.5 V	4.5 V		35		
	I_{CCL}	3005	89	0.0 V		5.5 V	5.5 V			5.5 V	5.5 V		"	"		5.5 V	5.5 V			5.5 V	5.5 V		5.5 V	V_{CC}		70	"
	I_{CCH}	"	90	0.0 V		0.0 V	0.0 V			0.0 V	0.0 V		"	"		0.0 V	0.0 V			0.0 V	0.0 V		"	"		50	"
	I_{CCZ}	"	91	5.5 V										"									"	"		61	"
2 T_C = +125°C	Same tests as subgroup 1 except V_{IC} tests are omitted.																										
3 T_C = -55°C	Same tests as subgroup 1 except V_{IC} tests are omitted.																										

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1 $\overline{CE}$	2 $\overline{Q_0}$	3 D_0	4 D_1	5 $\overline{Q_1}$	6 $\overline{Q_2}$	7 D_2	8 D_3	9 $\overline{Q_3}$	10 GND	11 LE	12 $\overline{Q_4}$	13 D_4	14 D_5	15 $\overline{Q_5}$	16 $\overline{Q_6}$	17 D_6	18 D_7	19 $\overline{Q_7}$	20 V_{CC}	Measured terminal	Test limits Min	Max	Unit
7 T_C = +25°C	Functional tests 1/2/3/	3014	92	B	L	A	B	H	L	A	B	H	GND	A	L	A	B	H	L	A	B	H	3/	All outputs			
			93		H	B	A	L	H	B	A	L		A	H	B	A	L	H	B	A	L					
			94			B	A			B	A			B		B	A			B	A						
			95			A	B	H	L	A	B	H		A	L	A	B	H	L	A	B	H					
			96		L	A	B	L	H	A	B	L		B	H	A	B	L	H	A	B	L					
			97		H	B	A	H	L	A	B	H		A	L	B	A	H	L	A	B	H					
			98		L	A	B	L	H	A	B	L		B	H	B	A	L	H	A	B	L					
			99			A	B			A	B			B		A	B			A	B						
			100			B	A			B	A			A		B	A			B	A						
8	Repeat subgroup 7 tests at T_C = +125°C and T_C = -55°C.																										
9 T_A = +25°C	t_{PLH1}	3003 Fig. 5	101	0.0 V	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND	2.7 V	OUT	IN	IN	OUT	OUT	IN	IN	OUT	5.0 V	IO_0 to IO_7	4.0	9.0	ns
			102																								
			103																								
			104																								
			105																								
			106																								
			107																								
			108																								
	t_{PHL1}		109		OUT	IN	IN	OUT	OUT	IN	IN	OUT			OUT	IN	IN	OUT	OUT	IN	IN	OUT		IO_0 to IO_7	3.0	17.0	
			110																								
			111																								
			112																								
			113																								
			114																								
			115																								
			116																								
	t_{PLH2}		117		OUT	0.0 V	0.0 V	OUT	OUT	0.0 V	0.0 V	OUT		IN	OUT	0.0 V	0.0 V	OUT	OUT	0.0 V	0.0 V	OUT		ILE to IO_0	5.0	11.0	
			118																					ILE to IO_1			
			119																					ILE to IO_2			
			120																					ILE to IO_3			
			121																					ILE to IO_4			
			122																					ILE to IO_5			
			123																					ILE to IO_6			
			124																					ILE to IO_7			
	t_{PHL2}		125		OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT			OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT		ILE to IO_0	3.0	17.0	
			126																					ILE to IO_1			
			127																					ILE to IO_2			
			128																					ILE to IO_3			
			129																					ILE to IO_4			
			130																					ILE to IO_5			
			131																					ILE to IO_6			
			132																					ILE to IO_7			

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,S,2 Test no.	1 $\overline{OE}$	2 $\overline{O_0}$	3 O_0	4 O_1	5 $\overline{O_1}$	6 $\overline{O_2}$	7 O_2	8 O_3	9 $\overline{O_3}$	10 GND	11 LE	12 $\overline{O_4}$	13 O_4	14 O_5	15 $\overline{O_5}$	16 $\overline{O_6}$	17 O_6	18 O_7	19 $\overline{O_7}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PHZ1}	3303 Fig. 5	133 134 135 136 137 138 139 140	In	OUT	0.0 V	3.0 V	OUT	OUT	0.0 V	3.0 V	OUT	GND	2.7 V	OUT	0.0 V	0.0 V	OUT	OUT	0.0 V	0.0 V	OUT	5.0 V	$\overline{O_0}$ to O_0 $\overline{O_1}$ to O_1 $\overline{O_2}$ to O_2 $\overline{O_3}$ to O_3 $\overline{O_4}$ to O_4 $\overline{O_5}$ to O_5 $\overline{O_6}$ to O_6 $\overline{O_7}$ to O_7	2.0	6.0	ns
	t_{PLZ1}		141 142 143 144 145 146 147 148		OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT		OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT			5.5		
	t_{PZH1}		149 150 151 152 153 154 155 156		OUT	0.0 V	3.0 V	OUT	OUT	3.0 V	0.0 V	OUT		OUT	0.0 V	0.0 V	OUT	OUT	0.0 V	3.0 V	OUT			10.0		
	t_{PZL1}		157 158 159 160 161 162 163 164		OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT		OUT	2.7 V	2.7 V	OUT	OUT	2.7 V	2.7 V	OUT	5.0 V		6.5		
10 T_C = +125°C	Same tests as subgroup 9 except use limits from table I.																										
11 T_C = -55°C	Same tests as subgroup 10 except T_C = -55°C.																										

1/ H = 2.5 V minimum, L = 0.5 V or GND.

2/ H ≥ 1.5 V, L ≤ 1.5 V

3/ Perform subgroup 7 sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

4/ I_{IL} limits (mA) min/max values for circuit shown:

Parameter	Circuits A	Circuits B	Circuits C
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,8,5 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OE}$	D_3	D_1	D_2	D_4	D_5	D_7	GND	LE	$\overline{D_4}$	$\overline{D_6}$	$\overline{D_5}$	$\overline{D_3}$	$\overline{D_1}$	$\overline{D_2}$	$\overline{D_0}$	V_{CC}						
1 T_C = +25°C	V_{OH}	3006	1	0.8 V						0.8 V		2.0 V	-1.0 mA									4.5 V		2.5		V
			2											-1.0 mA												
			3												-1.0 mA											
			4													-1.0 mA										
			5					0.8 V									-1.0 mA									
			6						0.8 V									-1.0 mA								
			7							0.8 V									-1.0 mA							
			8		0.8 V															-1.0 mA						
	V_{OL}	3007	9							2.0 V		1.1 V	20 mA									4.5 V		0.5		V
			10											20 mA												
			11												20 mA											
			12													20 mA										
			13					2.0 V									20 mA									
			14						2.0 V									20 mA								
			15							2.0 V									20 mA							
			16		2.0 V															20 mA						
	I_{IC}		17	-18 mA								1d mA												B0		-1.2	
			18		-18 mA																			B1			
			19			-18 mA																		B2			
			20				-18 mA																	B3			
			21					-1d mA																B4			
			22						-18 mA															B5			
			23							-1d mA														B6			
			24								-18 mA													B7			
			25										-18 mA											B8			
			26																					B9			
	I_{IH1}	3310	27	2.7 V								2.7 V										5.5 V		B0		20	µA
			28		2.7 V																			B1			
			29			2.7 V																		B2			
			30				2.7 V																	B3			
			31					2.7 V																B4			
			32						2.7 V															B5			
			33							2.7 V														B6			
			34								2.7 V													B7			
			35										2.7 V											B8			
			36									2.7 V												B9			
	I_{IH2}		37	7.0 V								7.0 V												B0		100	
			38		7.0 V																			B1			
			39			7.0 V																		B2			
			40				7.0 V																	B3			
			41					7.0 V																B4			
			42						7.0 V															B5			
			43							7.0 V														B6			
			44								7.0 V													B7			
			45										7.0 V											B8			
			46									7.0 V												B9			

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,8,5 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OE}$	O_0	O_1	O_2	O_3	O_4	O_5	O_6	O_7	GND	LE	$\overline{O_7}$	$\overline{O_6}$	$\overline{O_5}$	$\overline{O_4}$	$\overline{O_3}$	$\overline{O_2}$	$\overline{O_1}$	$\overline{O_0}$	V_{CC}				
1 T_C = +25°C	I_{IL}	3009	47 48 49 50 51 52 53 54 55 56	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	GND	0.5 V									5.5 V		1/	1/	mA
	I_{OS}	3011	57 58 59 60 61 62 63 64	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V		5.5 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V			-60	-150	mA
	I_{OZH}		65 66 67 68 69 70 71 72	2.0 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V			2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V				50	µA
	I_{OZL}		73 74 75 76 77 78 79 80		0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V			0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V			-50	µA
	I_{CCL} I_{CCZ}		81 82	0.0 V 5.5 V	5.5 V 0.0 V	5.5 V 0.0 V	5.5 V 0.0 V	5.5 V 0.0 V	5.5 V 0.0 V	5.5 V 0.0 V	5.5 V 0.0 V	5.5 V 0.0 V												V_{CC} V_{CC}		61 61	mA mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										

See footnotes at end of device type 01.

MIL-M-38510/346

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,8,S Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OE}$	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	GND	LE	$\overline{O_7}$	$\overline{O_6}$	$\overline{O_5}$	$\overline{O_4}$	$\overline{O_3}$	$\overline{O_2}$	$\overline{O_1}$	$\overline{O_0}$	V_{CC}				
7 T_C = +25°C	Functional 3/	3014	83	5	A	B	A	B	A	B	A	B	GND	A	H	L	L	L	H	L	H	L	1/	All outputs			
			84		B	A	B	A	B	A	B	A		A	L	H	L	L	L	H	L	H					
			85		B	A	B	A	B	A	B	A		B	L	H	L	H	L	H	L	H					
			86		A	B	A	B	A	B	A	B		B	L	H	L	H	L	H	L	H					
			87		A	B	A	B	A	B	A	B		A	H	L	H	L	H	L	H	L					
			88		B	A	B	A	B	A	B	A		A	L	H	L	H	L	H	L	H					
			89		A	B	A	B	A	B	A	B		A	H	L	H	L	H	L	H	L					
			90		A	B	A	B	A	B	A	B		B	H	L	H	L	H	L	H	L					
			91		B	A	B	A	B	A	B	A		B	H	L	H	L	H	L	H	L					
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																										
9 T_C = +25°C	t_{PLH1}	3003 Fig. 5	92	0.0 V	IN	IN	IN	IN	IN	IN	IN	IN	GND	2.7 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	5.0 V	IO_0 to $\overline{O_0}$	3.5	8.5	ns
			93																					IO_1 to $\overline{O_1}$			
			94																					IO_2 to $\overline{O_2}$			
			95																					IO_3 to $\overline{O_3}$			
			96																					IO_4 to $\overline{O_4}$			
			97																					IO_5 to $\overline{O_5}$			
			98																					IO_6 to $\overline{O_6}$			
			99																					IO_7 to $\overline{O_7}$			
	t_{PHL1}		100		IN	IN	IN	IN	IN	IN	IN	IN			OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		IO_0 to $\overline{O_0}$	2.5	6.5	
			101																					IO_1 to $\overline{O_1}$			
			102																					IO_2 to $\overline{O_2}$			
			103																					IO_3 to $\overline{O_3}$			
			104																					IO_4 to $\overline{O_4}$			
			105																					IO_5 to $\overline{O_5}$			
			106																					IO_6 to $\overline{O_6}$			
			107																					IO_7 to $\overline{O_7}$			
	t_{PLH2}		108		0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V		IN	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		LE to $\overline{O_0}$	4.5	9.5	
			109																					LE to $\overline{O_1}$			
			110																					LE to $\overline{O_2}$			
			111																					LE to $\overline{O_3}$			
			112																					LE to $\overline{O_4}$			
			113																					LE to $\overline{O_5}$			
			114																					LE to $\overline{O_6}$			
			115																					LE to $\overline{O_7}$			
	t_{PHL2}		116		2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V			OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		LE to $\overline{O_0}$	3.0	7.0	
			117																					LE to $\overline{O_1}$			
			118																					LE to $\overline{O_2}$			
			119																					LE to $\overline{O_3}$			
			120																					LE to $\overline{O_4}$			
			121																					LE to $\overline{O_5}$			
			122																					LE to $\overline{O_6}$			
			123																					LE to $\overline{O_7}$			
	t_{PHZ1}	3003	124	IN	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V		2.7 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		IO_0 to $\overline{O_0}$	1.5	5.5	
			125																					IO_1 to $\overline{O_1}$			
			126																					IO_2 to $\overline{O_2}$			
			127																					IO_3 to $\overline{O_3}$			
			128																					IO_4 to $\overline{O_4}$			
			129																					IO_5 to $\overline{O_5}$			
			130																					IO_6 to $\overline{O_6}$			
			131																					IO_7 to $\overline{O_7}$			
	t_{PLZ1}		132		2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V			OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		IO_0 to $\overline{O_0}$	1.5	5.5	
			133																					IO_1 to $\overline{O_1}$			
			134																					IO_2 to $\overline{O_2}$			
			135																					IO_3 to $\overline{O_3}$			
			136																					IO_4 to $\overline{O_4}$			
			137																					IO_5 to $\overline{O_5}$			
			138																					IO_6 to $\overline{O_6}$			
			139																					IO_7 to $\overline{O_7}$			

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S test no.	1 $\overline{OE}$	2 D_3	3 D_1	4 D_2	5 D_3	6 D_4	7 D_5	8 D_6	9 D_7	10 GND	11 LE	12 $\overline{O_7}$	13 $\overline{O_6}$	14 $\overline{O_5}$	15 $\overline{O_4}$	16 $\overline{O_3}$	17 $\overline{O_2}$	18 $\overline{O_1}$	19 $\overline{O_0}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PZH1}		140	IN	0.0 V								GND	2.7 V									5.0 V	D_0 to O_0	2.0	7.5	ns
			141			0.0 V																OUT		D_1 to O_1			
			142				0.0 V														OUT			D_2 to O_2			
			143					0.0 V												OUT				D_3 to O_3			
			144						0.0 V										OUT					D_4 to O_4			
			145							0.0 V								OUT						D_5 to O_5			
			146								0.0 V						OUT							D_6 to O_6			
			147									0.0 V			OUT									D_7 to O_7			
	t_{PZL1}		148		2.7 V																			D_0 to O_0	3.0	8.5	
			149			2.7 V																OUT		D_1 to O_1			
			150				2.7 V														OUT			D_2 to O_2			
			151					2.7 V												OUT				D_3 to O_3			
			152						2.7 V										OUT					D_4 to O_4			
			153							2.7 V								OUT						D_5 to O_5			
			154								2.7 V						OUT							D_6 to O_6			
			155									2.7 V			OUT									D_7 to O_7			
10	Same tests as subgroup 9 except use T_C = +125°C and use limits from table I.																										
11	Same tests as subgroup 10 except T_C = -55°C																										

Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuit A	Circuit B	Circuit C
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60

A = 2.5 V minimum; B = 0.3 V or GND, H ≥ 2.5 L ≤ 0.5 V.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OE}$	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	GND	$\overline{LE}$	Q_7	Q_6	Q_5	Q_4	Q_3	Q_2	Q_1	Q_0	V_{CC}				
T_C = +25°C	V_{OH}	3006	1	0.8 V								2.0 V		2.0 V	-1.0 mA								4.5 V	Q_7	2.5		V
			2								2.0 V					-1.0 mA								Q_6			
			3							2.0 V							-1.0 mA							Q_5			
			4						2.0 V									-1.0 mA						Q_4			
			5					2.0 V											-1.0 mA					Q_3			
			6				2.0 V													-1.0 mA				Q_2			
			7			2.0 V															-1.0 mA			Q_1			
			8		2.0 V																	-1.0 mA		Q_0			
	V_{OL}	3007	9											1.1 V	20 mA									Q_7		0.5	V
			10									0.8 V				20 mA								Q_6			
			11								0.8 V						20 mA							Q_5			
			12						0.8 V									20 mA						Q_4			
			13					0.8 V		0.8 V									20 mA					Q_3			
			14				0.8 V													20 mA				Q_2			
			15			0.8 V															20 mA			Q_1			
			16		0.8 V																	20 mA		Q_0			
	V_{IC}		17	-18 mA										-18 mA										$\overline{OE}$		-1.2	V
			18		-18 mA																			D_0			
			19			-18 mA																		D_1			
			20				-18 mA																	D_2			
			21					-18 mA																D_3			
			22						-18 mA															D_4			
			23							-18 mA														D_5			
			24								-18 mA													D_6			
			25									-18 mA												D_7			
			26											-18 mA										$\overline{LE}$			
	I_{IH1}	3010	27	2.7 V																			5.5 V	$\overline{OE}$		20	μA
			28		2.7 V																			D_0			
			29			2.7 V																		D_1			
			30				2.7 V																	D_2			
			31					2.7 V																D_3			
			32						2.7 V															D_4			
			33							2.7 V														D_5			
			34								2.7 V													D_6			
			35									2.7 V												D_7			
			36											2.7 V										$\overline{LE}$			
	I_{IH2}		37	7.0 V																				$\overline{OE}$		100	
			38		7.0 V																			D_0			
			39			7.0 V																		D_1			
			40				7.0 V																	D_2			
			41					7.0 V																D_3			
			42						7.0 V															D_4			
			43							7.0 V														D_5			
			44								7.0 V													D_6			
			45									7.0 V												D_7			
			46											7.0 V										$\overline{LE}$			

See footnotes at end of device type 04.

MIL-M-38510/34AA

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,8,5 Test no.	1 $\overline{OE}$	2 O_0	3 O_1	4 O_2	5 O_3	6 O_4	7 O_5	8 O_6	9 O_7	10 GND	11 LE	12 O_7	13 O_6	14 O_5	15 O_4	16 O_3	17 O_2	18 O_1	19 O_0	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IL}	3009	47 48 49 50 51 52 53 54 55 56	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	GND	0.5 V									5.5 V	$\overline{OE}$ O_0 O_1 O_2 O_3 O_4 O_5 O_6 O_7 LE	1/	1/	mA
	I_{OS}	3011	57 58 59 60 61 62 63 64	0.0 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V		O_7 O_6 O_5 O_4 O_3 O_2 O_1 O_0	-60	-150	mA
	I_{OZH}		65 66 67 68 69 70 71 72	2.0 V	10.0 V	0.0 V	0.0 V	0.0 V	0.0 V	10.0 V	0.0 V	0.0 V			2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V		O_7 O_6 O_5 O_4 O_3 O_2 O_1 O_0		50	µA
	I_{OZL}		73 74 75 76 77 78 79 80		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V			0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V		O_7 O_6 O_5 O_4 O_3 O_2 O_1 O_0		-50	
	I_{CCL} I_{CCZ}		81 82	0.0 V 5.5 V	10.0 V 10.0 V	10.0 V 10.0 V	10.0 V 10.0 V	10.0 V 10.0 V	10.0 V 10.0 V	10.0 V 10.0 V	10.0 V 10.0 V	10.0 V 10.0 V											V_{CC} V_{CC}		55 55	mA mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{CE}$	O_0	O_1	O_2	O_3	O_4	O_5	O_6	O_7	GND	LE	O_7	O_6	O_5	O_4	O_3	O_2	O_1	O_0	V_{CC}				
7 T_C = +25°C	Functional 3/	3014	83	9	9	A	B	A	B	A	B	A	GND	A	H	L	H	L	H	L	H	L	1/	All outputs			
		"	84	"	A	B	A	B	A	B	A	B	"	A	L	H	L	H	L	H	L	H	"	"			
		"	85	"	A	B	A	B	A	B	A	B	"	B	L	H	L	H	L	H	L	H	"	"			
		"	86	"	B	A	B	A	B	A	B	A	"	B	L	H	L	H	L	H	L	H	"	"			
		"	87	"	B	A	B	A	B	A	B	A	"	A	H	L	H	L	H	L	H	L	"	"			
		"	88	"	B	A	B	A	B	A	B	A	"	A	L	H	L	H	L	H	L	H	"	"			
		"	89	"	B	A	B	A	B	A	B	A	"	A	H	L	H	L	H	L	H	L	"	"			
		"	90	"	B	A	B	A	B	A	B	A	"	B	H	L	H	L	H	L	H	L	"	"			
		"	91	"	B	A	B	A	B	A	B	A	"	B	H	L	H	L	H	L	H	L	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																										

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R,S Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{PE}$	D_3	D_1	D_2	D_3	D_4	D_5	D_6	D_7	GND	LE	Q_7	Q_6	Q_5	Q_4	Q_3	Q_2	Q_1	Q_0	V_{CC}				
9 T_C = +25°C	t_{PLH1}	3003 Fig. 5	92 93 94 95 96 97 98 99	0.0 V	IN	IN	IN	IN	IN	IN	IN	IN	GND	2.7 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	5.0 V	O_0 to Q_0 O_1 to Q_1 O_2 to Q_2 O_3 to Q_3 O_4 to Q_4 O_5 to Q_5 O_6 to Q_6 O_7 to Q_7	1.0	7.0	ns
	t_{PHL1}		100 101 102 103 104 105 106 107		IN	IN	IN	IN	IN	IN	IN	IN			OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		O_0 to Q_0 O_1 to Q_1 O_2 to Q_2 O_3 to Q_3 O_4 to Q_4 O_5 to Q_5 O_6 to Q_6 O_7 to Q_7	2.0	8.0	
	t_{PLH2}		108 109 110 111 112 113 114 115		2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V		IN	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		ILE to Q_0 ILE to Q_1 ILE to Q_2 ILE to Q_3 ILE to Q_4 ILE to Q_5 ILE to Q_6 ILE to Q_7	5.0	11.0	
	t_{PHL2}		116 117 118 119 120 121 122 123	IN	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V			OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		ILE to Q_0 ILE to Q_1 ILE to Q_2 ILE to Q_3 ILE to Q_4 ILE to Q_5 ILE to Q_6 ILE to Q_7	3.0	7.0	
	t_{PHZ1}	3003	124 125 126 127 128 129 130 131		2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V		2.7	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		$\overline{OE}$ to Q_0 $\overline{OE}$ to Q_1 $\overline{OE}$ to Q_2 $\overline{OE}$ to Q_3 $\overline{OE}$ to Q_4 $\overline{OE}$ to Q_5 $\overline{OE}$ to Q_6 $\overline{OE}$ to Q_7	1.5	5.5	
	t_{PLZ1}		132 133 134 135 136 137 138 139		0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V			OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		$\overline{OE}$ to Q_0 $\overline{OE}$ to Q_1 $\overline{OE}$ to Q_2 $\overline{OE}$ to Q_3 $\overline{OE}$ to Q_4 $\overline{OE}$ to Q_5 $\overline{OE}$ to Q_6 $\overline{OE}$ to Q_7			

See footnotes at end of device type 04.

MIL-M-38510/36B

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, 8, 5 Test no.	1 $\overline{OE}$	2 O_0	3 O_1	4 O_2	5 O_3	6 O_4	7 O_5	8 O_6	9 O_7	10 GND	11 LE	12 O_7	13 O_6	14 O_5	15 O_4	16 O_3	17 O_2	18 O_1	19 O_0	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PZH1}		140	IN	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	GND	2.7	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	5.0 V	$\overline{OE}$ to O_0	2.0	9.0	ns
			141																				$\overline{OE}$ to O_1			
			142																				$\overline{OE}$ to O_2			
			143																				$\overline{OE}$ to O_3			
			144																				$\overline{OE}$ to O_4			
			145																				$\overline{OE}$ to O_5			
			146																				$\overline{OE}$ to O_6			
			147																				$\overline{OE}$ to O_7			
	t_{PZL1}		148		0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V			OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT		$\overline{OE}$ to O_0		8.5	
			149																				$\overline{OE}$ to O_1			
			150																				$\overline{OE}$ to O_2			
			151																				$\overline{OE}$ to O_3			
			152																				$\overline{OE}$ to O_4			
			153																				$\overline{OE}$ to O_5			
			154																				$\overline{OE}$ to O_6			
			155																				$\overline{OE}$ to O_7			
10	Same tests as subgroup 9 except T_C = +125 and use limits from table I.																									
11	Same tests as subgroup 10 except T_C = -55°C																									

1/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

2/ I_{IL} limits (mA) min/max values for circuit shown:

Parameter	Circuits A	Circuits B	Circuits C
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60

A = 2.5 V minimum; B = 0.5 V or GND, H ≥ 2.5 L ≤ 0.5 V.

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e. Requirements for failure analysis (including required test condition of method 5005 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - Ground zero voltage potential.
I_{IN} - - - - - - - - - - - - - Current flowing into an input terminal.
V_{IN} - - - - - - - - - - - - - Voltage level at an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F373
02	54F533
03	54F563
04	54F573

6.6 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit		
	A	B	C
	Fairchild Semiconductor	Motorola	Signetics
01	X	X	
02	X	X	
03	X		
04	X		

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/346B	Microcircuits, Digital, Bipolar, Advanced Schottky TTL, Transparent and Octal D-Type Latches, Cascadable, Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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